Supplement to "The maximum size of a blackout that keeps every lattice rectangle identifiable"

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This supplement describes the SAT encodings behind the upper bounds of the theorem on small heights, proves that they are faithful, and lists the instances, the certificates, and the formal proofs. We refer to results of the paper by their names. Paths are relative to the root of the repository https://github.com/apemm/rectangle-blackouts.

S1 The SAT encodings

Let $C(R)$ be the corner set of a rectangle R. A blackout S is valid if and only if it contains no set $C(R_1)\triangle C(R_2)$ with $R_1 \neq R_2$ (the lemma on presentations in Section 2 of the paper).

Encoding A has a variable for each grid point and a clause for each inclusion-minimal set of the form $C(R_1)\triangle C(R_2)$ with $R_1 \neq R_2$. The clause says that this set is not fully black.

Encoding B has such clauses only for the pairs of rectangles that share a corner. It also has a variable y_R for each rectangle R, which is forced to be true if R is fully black, and it allows at most 1 true y_R.

A *structured* instance on 6 rows fixes one row graph from the equality structure theorem (a $K_{3,3}$, for $m+9$ points) or from the lemma on 6 rows and $m+8$ points ($K_{3,3}$ minus an edge, $K_{2,4}$, or $K_{3,3}$ with 1 empty column). Each column chooses a type, which is a single row, an edge of the graph, or the empty set in the last case. Each edge is used once, and the empty type is used once in the last case. The instance keeps those clauses of encoding B that involve a tilted rectangle and whose set can be fully black for these types. The size of the blackout is fixed by a sequential counter [5] or a totalizer [1].

Lemma S1. *Let S be a blackout of the $h \times m$ grid.*

(a) S is valid if and only if it contains no inclusion-minimal set of the form $C(R_1)\triangle C(R_2)$ with $R_1 \neq R_2$.

(b) S is valid if and only if at most 1 rectangle is fully black and S contains no set $C(R_1)\triangle C(R_2)$ for distinct rectangles R_1, R_2 that share a corner.

(c) Let $h = 6$ and $m \geq 3$, and let S be valid with $m+9$ or $m+8$ points. Then for one of the row graphs above, the choice of the black rows of column i as the type of column i extends to a satisfying assignment of the structured instance.

Proof. (a) Every set $C(R_1)\triangle C(R_2)$ contains an inclusion-minimal one. (b) For rectangles without a common corner, the set $C(R_1)\triangle C(R_2)$ is $C(R_1) \cup C(R_2)$, which is black exactly when both rectangles are fully black. (c) The equality structure theorem or the lemma on $m+8$ points gives the row graph, and the black rows of every column form an allowed type. Set y_R to true exactly for the fully black rectangles. By (b) this satisfies encoding B, and the structured instance has a subset of its clauses. □

Hence the point variables of encodings A and B describe exactly the valid blackouts of the chosen size. On 6 rows, $f(6, m) \leq m + 8$ once the 10 structured instances with $m + 9$ points are unsatisfiable, and $f(6, m) \leq m + 7$ once the further 115 instances with $m + 8$ points are.

S2 Instances and certificates

CaDiCaL [2] produced an LRAT proof for every unsatisfiable instance in Table S1. Each of the 739 proofs was accepted by 2 checkers, the program `lrat-check` of drat-trim [6, 3] and the LRAT checker of the Lean core library. The proofs are large and are not stored. Their hashes, the hashes of the CNF files, and the results of both checkers are in `jis/data/unsat_certificates.json` and `jis/data/six_row_excess8_certificates.json`. The pipeline in `jis/unsat` recreates the proofs. Two separately written generators produce the same clauses for the instances of encoding A and for the 70 structured instances with $m + 9$ points. Encoding B also finds all 345 structured instances with $m + 8$ points unsatisfiable. SAT enumeration uses PySAT [4].

bound	instances	proofs
$f(5, 6) < 12$, $f(5, 7) < 13$	encoding A, 1 instance each	2
$f(6, 6) < 13$, $f(6, 7) < 14$	encoding A, 1 instance each	2
$f(6, 8) < 16$	encoding A, split by the first column	64
$f(7, 7) < 15$	encoding A, split by the first column	128
$f(7, 8) < 17$	encoding A, split by the first column	128
$f(6, m) < m + 9$ for $8 \leq m \leq 14$	structured, 10 row graphs for each width	70
$f(6, m) < m + 8$ for $9 \leq m \leq 11$	structured, 115 row graphs for each width	345

Table S1: The unsatisfiable instances.

The 72 and 1440 candidates with $m + 4$ points on the 4×4 and 4×5 grids are searched by `verify_narrow.py`, which finds 2 rectangles with the same presentation in each of them.

S3 Explicit blackouts

The blackouts and test boards of Section 6 of the paper are checked from the definition by the scripts in `jis/scripts` and by `papers/extremal/scripts/check_constructions.py`. The second script lists the tilted rectangles through their leftmost corner, compares all presentations with at most 2 points, and looks for an axis-aligned rectangle with the same presentation. It also checks the explicit construction with spacing $W_h + 1$ for $3 \leq h \leq 8$, and blackouts built as in Section 5 for $4 \leq h \leq 14$ and $h = 16$. In these, each double gets the leftmost column that is not bad, where the conditions (B1), (B2), and (B3) are tested as defined. At $h = 16$, all 3 blackouts built in this way are valid, and all 5 blackouts built with (B1) alone are not valid. For $h = 20$ and $h = 24$ the script only computes the columns and prints the width. The script runs in about 4 minutes.

S4 Formal proofs

The Lean files are in `lean/KageyProblems/Problem001`. They contain formal proofs of the lemma on presentations, of the lemma that 3 corners determine a rectangle, of the span bound, of the row bound, of the finite collar theorem, and of the alternating seas proposition under the assumption that the doubles are $2W_h + 2$ columns apart (`GeneralLocal.lean`

and `GeneralTheorem.lean`). This gives $f(h,m) = m + \lfloor h^2/4 \rfloor$ for every $h \geq 3$ and $m \geq q_h + (q_h - 1)(2W_h + 1)$. Validity tests of S_0, S_8, the 6×8 and 6×15 blackouts, and the 5×13 and 6×23 test boards are checked by the Lean kernel. Together with the formal row bound and collar theorem, this proves $f(5,m) = m + 6$ for $m \geq 8$, $f(6,m) = m + 9$ for $m \geq 15$, $f(5,5) \geq 10$, and $f(6,8) \geq 15$ in Lean.

Not formalized are the spacing $W_h + 1$, the threshold theorem and its corollary, the equality structure theorem, the lemma on 6 rows and $m + 8$ points, the blackouts on 7 rows, and Lemma S1.

References

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